

CSS-Assignment 1
Date of Assignment-28th Feb'26
Date of Submission-9th Mar'26

1. GSM needs to send this breaking news securely:

"GLOBAL MARKETS SHOW STRONG RECOVERY"

Encrypt the message using Caesar Cipher with key = 5.

Show:

- Formula used
- Step-by-step shifting
- Final cipher text

2. An encrypted message intercepted from a rival newsroom:

"YMJHJXF WHNUMJWNXXNRUQJ"

It is known that Caesar cipher was used with key = 5.

Decrypt the message and write the original statement.

3. A secure message must be sent by Election Security Bulletin:

"ELECTION RESULTS VERIFIED"

Use Affine Cipher with:

$$a = 5$$

$$b = 8$$

Encrypt the message and show calculations.

4. An encrypted diplomatic cable update reads:

"ZEBBW"

It was encrypted using Affine Cipher with:

$$a = 5$$

$$b = 8$$

Decrypt the message.

5. Encrypt plaintext **"HELP"** using Hill Cipher with key matrix:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Steps required:

- Convert letters to numbers
- Divide into digraphs
- Matrix multiplication
- Mod 26 operation
- Final ciphertext

6. Message to encrypt:

"CLIMATE SUMMIT AGREEMENT SIGNED"

Keyword: GREEN

Encrypt using Vigenère Cipher.

Show: Final ciphertext

7. Encrypted message:

"RIJVSUYVJN"

Keyword: KEY

Decrypt the message using Vigenère Cipher

8. Message: **"MOON LANDING SUCCESS"**

Key (same length): "XMCKLQWERTYPLMNBVD"

Encrypt using Vernam cipher.

9. A defense command needs to securely transmit information about an upcoming air mission using the One-Time Pad (OTP) technique.

Plaintext:

FIGHTERS ATTACK AT NOON

Random Key (same length, truly random):

ZMXNBLKJHGFDSAPOIUYTRE

Encrypt the plaintext using the One-Time Pad (OTP) method.

10. Keyless Cipher (Rail Fence)

Encrypt:

"WORLD LEADERS MEET TODAY"

Using Rail Fence Cipher with 3 rails.

11. Keyed Cipher (Columnar Transposition)

Encrypt:

"INTERNATIONAL PEACE TALKS"

Using key: MEDIA

Show:

- Column arrangement
- Final ciphertext

12. Breaking diplomatic alert:

"CEASE FIRE AGREED"

Using keyword:

"DIPLOMACY"

Tasks:

- Prepare plaintext (remove spaces, handle repeated letters, add filler if needed).
- Divide into digraphs.
- Apply Playfair rules.
- Show final ciphertext.

13. An intercepted ciphertext reads:

"BMNDZBXDKYBEJVDMUIXMMNUVIF"

It was encrypted using Playfair cipher with keyword:

"MONARCHY"

Tasks:

- Construct the Playfair matrix.
- Decrypt the message.

14. A secure communication server is generating encryption keys.

Two large numbers are selected: 414 and 662

- Before generating a key pair, the system must verify whether the numbers are coprime.
- Use the Euclidean Algorithm to find $\gcd(414, 662)$.
- Show all steps clearly.
- Explain why checking gcd is necessary before encryption.
- If $\gcd \neq 1$, what risk does it pose in cryptographic systems?

15. A journalist encrypts data using a public key exponent:

$e = 17$

$\phi(n) = 3120$

Use the Extended Euclidean Algorithm to find d .

16. A security engineer mistakenly selects:

$$e = 12$$

$$\phi(n) = 30$$

- Use Euclidean Algorithm to check if modular inverse exists.
- Can Extended Euclidean Algorithm find d ?
- Explain mathematically why modular inverse does or does not exist.
- What condition must hold for modular inverse existence?

17. A secure server chooses:

$$p = 13$$

$$q = 19$$

To generate keys, the system must compute $\phi(n)$.

- Compute n .
- Compute $\phi(n)$.
- Explain why $\phi(n) = (p-1)(q-1)$ works only when p and q are prime.
- What happens if one of the numbers is not prime?

18. A network system uses modulo 20 arithmetic.

The system must count how many integers less than 20 are coprime with 20.

- Compute $\phi(20)$.
- List all integers that are coprime with 20.
- Verify your result using prime factorization formula.
- Why is totient function important in encryption systems?
- If n is prime, what is $\phi(n)$?

19. Apply **Fermat's Little Theorem** to compute the remainder when 10^{2025} is divided by 19.

20. A secure system uses the **Autokey cipher** for encryption.

Encrypt the plaintext: "**CYBER ATTACK TARGETS BANKS**"
using the initial key: K